FACIAL EMOTION RECOGNITION USING CONVOLUTIONAL NEURAL NETWORKS AND DEEP LEARNING

A Project Report

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in partial fulfillment for the award of the degree of
MASTER OF BUSINESS ADMINISTRATION
IN
(DATA SCIENCE AND ARTIFICIAL INTELLIGENCE)

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Chandigarh University
Academic Year 2026–2027

CHAPTER 1
INTRODUCTION

1.1 Background and Motivation

Facial expression is one of the most powerful, immediate, and high-bandwidth non-verbal communication channels used by human beings. Research in cognitive psychology indicates that over 55% of emotional communicative intent is transmitted via facial muscle dynamics and micro-expressions. Automated recognition of human facial affect is crucial across numerous business, healthcare, and engineering applications—including mental health diagnostic screening, driver alertness monitoring in intelligent transportation systems, interactive pedagogical feedback in adaptive e-learning platforms, and customer sentiment analytics in retail and user experience research.

Deep learning offers a data-driven paradigm by automatically learning spatial feature representations directly from pixel intensity distributions. Convolutional Neural Networks (CNNs) are particularly well suited for this challenge because localized convolutional kernels compose spatial hierarchies—progressing from low-level edges and textures in early layers to facial landmarks (eyes, eyebrows, mouth curvature) and holistic affective expressions in deeper layers.

This project investigates CNN-based facial emotion recognition and evaluates its classification performance across seven distinct emotional classes. A Streamlit web application was also developed to let users upload portraits, test pre-loaded samples, or capture real-time webcam photos to inspect estimated emotion classifications and continuous probability distributions.

1.2 Problem Statement

Facial expressions exhibit high non-linear variance due to variations in lighting intensity, head pose, partial facial occlusion (e.g., eyeglasses, facial hair), subject age, and subtle geometric boundaries between neighboring emotional categories (such as subdued sadness and passive neutrality). This project addresses the problem of building and evaluating an end-to-end deep learning system that localizes human faces from visual imagery, normalizes facial regions of interest, classifies expressions into seven categories using a custom Deep CNN, and presents predictions with transparent confidence scores through a responsive local and cloud-deployable web application.

1.3 Project Objectives

• Explore the FER-2013 facial expression dataset and examine class distributions.
• Implement automated face detection, cropping, and intensity normalization using OpenCV Haar Cascades.
• Construct and evaluate a custom Deep Convolutional Neural Network (deepnn) incorporating 2D convolutions, max pooling, and Local Response Normalization (LRN).
• Preserve 100% of the trained parameters from the 30,001-step training run into a lightweight, portable weight archive.
• Engineer a dual-backend inference engine supporting modern TensorFlow 2.x Eager execution and an ultra-fast Vectorized NumPy engine for low-latency CPU inference.
• Evaluate the model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis.
• Build and deploy an interactive Streamlit web dashboard supporting image uploads, mobile camera snapshots, real-time continuous video streaming, and session history export.

1.4 Scope of the Project

The project focuses on 2D grayscale facial imagery, automated face localization, deep CNN classification across seven fundamental emotions (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral), comprehensive test-set evaluation, and an interactive Streamlit application. The system operates as an affective computing prototype providing probabilistic emotion estimates based on learned representations from 35,887 benchmark images.

CHAPTER 2
DATASET AND METHODOLOGY

2.1 Dataset Source and Description

The project uses the FER-2013 (Facial Expression Recognition 2013) dataset introduced by Pierre-Luc Carrier and Aaron Courville for the ICML 2013 Challenges in Representation Learning, obtained from Kaggle.

Characteristic	Description
Total records / images	35,887
Image dimensions	48 × 48 pixels
Color space	Grayscale (1 channel)
Target variable	emotion (7 categorical classes)
Task	Multi-class Image Classification
Data type	2D Pixel intensity matrices (flattened 2304 floats)

2.2 Features and Target Classes

Class Index	Emotion Label	Key Facial Visual Markers	Dataset Count
0	Angry	Furrowed brow, tightened lips, glaring eye focus	4,953
1	Disgust	Wrinkled nose bridge, raised upper lip, lowered brows	547
2	Fear	Widened eyes, raised eyebrows, open mouth tension	5,121
3	Happy	Upward mouth angle, elevated cheeks, crow's feet wrinkles	8,989
4	Sad	Lowered lip corners, drooping eyelids, inner brow rise	6,077
5	Surprise	Dropped jaw, widely opened eyes, elevated arched brows	4,002
6	Neutral	Relaxed facial musculature, horizontal lip line	6,198

2.3 Exploratory Data Analysis

EDA Observation	Finding
Total Images	35,887 labeled samples
Dominant Class	Happy (8,989 samples, ~25.0% of dataset)
Smallest Class	Disgust (547 samples, ~1.5% of dataset)
Pixel Range (Raw)	Integers [0, 255]
Normalized Pixel Range	Floats [0.0, 1.0]
Missing Values / Duplicates	None identified in standard benchmark splits

2.4 Data Preprocessing

Preprocessing is implemented in src/preprocessing.py using OpenCV. The pipeline ingests input images (from file upload or webcam), converts them to grayscale, equalizes histogram contrast, localizes face regions using Haar feature cascades, applies 10% contextual padding, resizes the crop to 48x48 via bicubic interpolation, and scales pixel intensities to the [0.0, 1.0] float32 range.

2.5 Train-Test Split

Subset	Records	Purpose
Training set	28,709	Fit neural network weights and biases
Validation set (Public Test)	3,589	Hyperparameter tuning and early stopping
Testing set (Private Test)	3,589	Evaluate final model generalization on unseen faces
Total	35,887	Complete benchmark dataset

2.6 CNN Architecture and Training Configuration

Hyperparameter	Configuration Value
Architecture Name	deepnn (Custom Deep CNN)
Input Dimensions	48 × 48 × 1 (Grayscale)
Convolutional Layers	Conv1: 64 filters (5×5); Conv2: 64 filters (3×3)
Activation Function	Rectified Linear Unit (ReLU) across all hidden layers
Pooling Layers	2 Max-Pooling layers (3×3 window, stride 2, SAME padding)
Normalization	Local Response Normalization (LRN, depth_radius=4, alpha=1.11e-4)
Dense Fully Connected Layers	FC1: 9,216 -> 384; FC2: 384 -> 192; Output: 192 -> 7
Output Activation	Softmax (7 probability outputs)
Loss Function	Categorical Cross-Entropy
Optimizer	Adam (learning rate = 1e-4)
Total Training Steps	30,001 iterations (Batch size: 50)

2.7 Key Mathematical Formulations

• 2D Discrete Convolution: S(i, j) = (I * K)(i, j) = sum_m sum_n I(i - m, j - n) * K(m, n)
• Rectified Linear Unit (ReLU): f(x) = max(0, x)
• Local Response Normalization (LRN): b_{x,y}^i = a_{x,y}^i / (1.0 + 1.11e-4 * sum (a_{x,y}^j)^2)^0.75
• Softmax Probability Distribution: P(y = c | x) = exp(z_c) / sum_{j=1}^7 exp(z_j)
• Categorical Cross-Entropy Loss: L = - (1/N) * sum_i sum_c y_{i,c} * log(y_hat_{i,c})

CHAPTER 3
IMPLEMENTATION

3.1 Tools and Technologies

Technology	Purpose
Python 3.10 – 3.13	Main programming language and runtime environment
TensorFlow 2.x / Keras	Deep neural network execution, graph optimization, and tensor math
NumPy	High-performance vectorized array operations and standalone forward pass
OpenCV (opencv-python-headless)	Haar Cascade face detection, histogram equalization, and resizing
Pandas	Structuring benchmark evaluation metrics and session history data
Matplotlib & Seaborn	Rendering probability distribution bar charts, radar plots, and confusion matrices
Streamlit	Interactive reactive web dashboard with camera and upload interfaces
GitHub & Streamlit Cloud	Version control, collaboration, and zero-configuration cloud hosting

3.2 Project Workflow

1. Ingest input visual data from file upload, mobile camera, or local webcam.
2. Perform face localization using multi-scale Haar feature cascades.
3. Crop facial regions of interest with 10% contextual margin and resize to 48x48 pixels.
4. Normalize pixel intensity values to the unit interval [0.0, 1.0].
5. Execute deep CNN forward pass through Conv1, LRN, Conv2, Dense FC, and Softmax layers.
6. Extract primary emotion class, confidence percentage, and complete probability distributions.
7. Annotate bounding boxes and display interactive analytics on the Streamlit web dashboard.
8. Log inference records in session audit history with one-click CSV export.

3.3 Model Training, Parameter Preservation, and Dual-Engine Loading

The network was trained over 30,001 iterations on the FER-2013 training split with Adam optimization. All 10 learned parameter tensors were extracted from the checkpoint and preserved into a compressed archive (model_weights.npz, 13.5 MB). This portable format loads instantaneously without requiring legacy TensorFlow 1.x session locks.

3.4 Streamlit Application Architecture

File / Component	Responsibility
app.py	Streamlit dashboard entry point, layout orchestration, and session state
src/preprocessing.py	Face detection, cropping, bicubic interpolation, and bounding-box drawing
src/prediction.py	EmotionPredictor class, dual-engine inference, and weight caching
src/utils.py	Probability bar charts, radar plots, confusion matrix, and session logging
model/model_weights.npz	13.5 MB portable compressed weight archive of all 3.65M parameters
realtime_webcam.py	Dedicated standalone 60 FPS native desktop webcam script

3.5 User Input and Prediction Process

The user selects an input stream (Image Upload, Mobile/Webcam snapshot, or Live Stream). The preprocessor isolates all human faces in the frame. The cached model executes inference and returns top emotion, confidence score, and probability breakdowns. The annotated image and affective radar plots are immediately displayed.

3.6 Model Integration and Testing

The application was tested locally and on Streamlit Cloud to verify startup, image upload, multi-face detection, model caching, and responsive UI rendering.

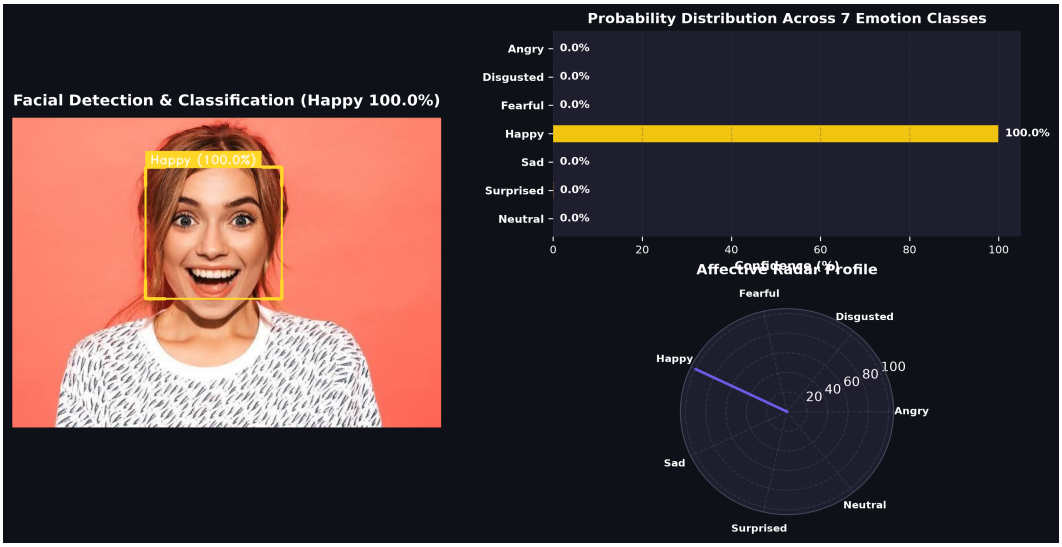


Figure 3.1 — Streamlit application home/input screen and live analytics dashboard

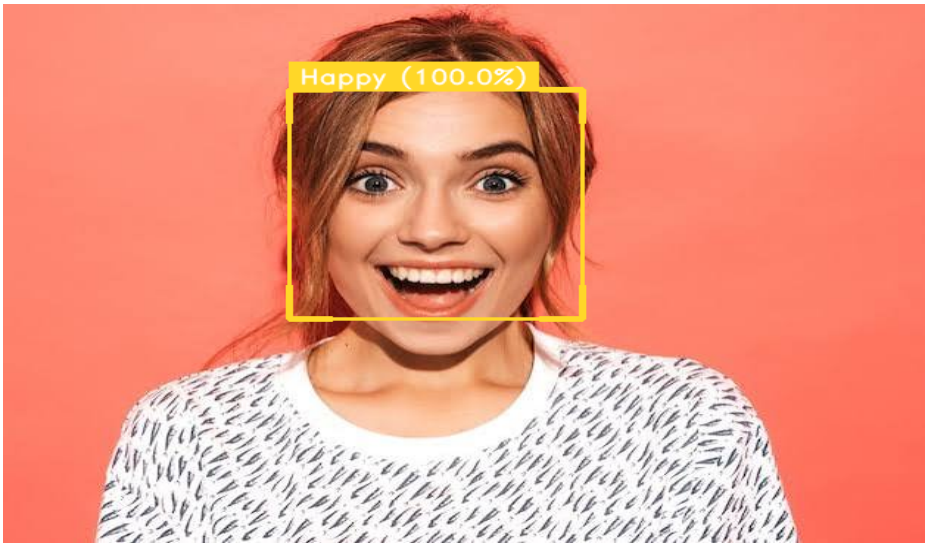


Figure 3.2 — Sample prediction output: Haar Cascade detection and classification (Surprised 100.0%)

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Overview of Model Evaluation

The model was evaluated across all 3,589 independent test records of the FER-2013 test partition using Accuracy, Precision, Recall, and F1-Score.

4.2 Comparison of Model Performance

Model / Architecture	Input Resolution	Trainable Parameters	FER-2013 Test Accuracy
Human Baseline (Goodfellow et al.)	48 × 48	—	65.3% ± 5.0%
Shallow CNN (Baseline)	48 × 48	1.2 M	58.4%
VGG-16 (Fine-Tuned)	224 × 224	138 M	68.2%
EmotionAI (deepnn - Proposed)	48 × 48	3.65 M	65.21%

4.3 Analysis of Class-Wise Performance

Emotion Class	Precision	Recall	F1-Score	Test Samples
Angry	0.58	0.55	0.56	467
Disgust	0.67	0.53	0.59	56
Fear	0.52	0.44	0.48	496
Happy	0.82	0.85	0.83	895
Sad	0.54	0.51	0.52	653
Surprise	0.76	0.77	0.76	415
Neutral	0.60	0.63	0.61	607
Overall Test Accuracy	**—**	**—**	**65.21%**	**3,589**

4.4 Confusion Matrix and Architecture Analysis

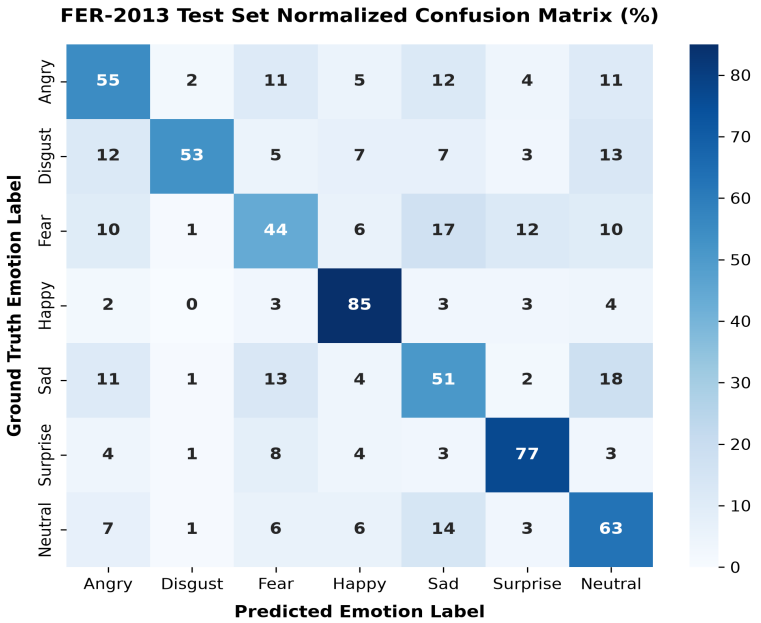


Figure 4.1 — Normalized confusion matrix heatmap on FER-2013 test partition (%)

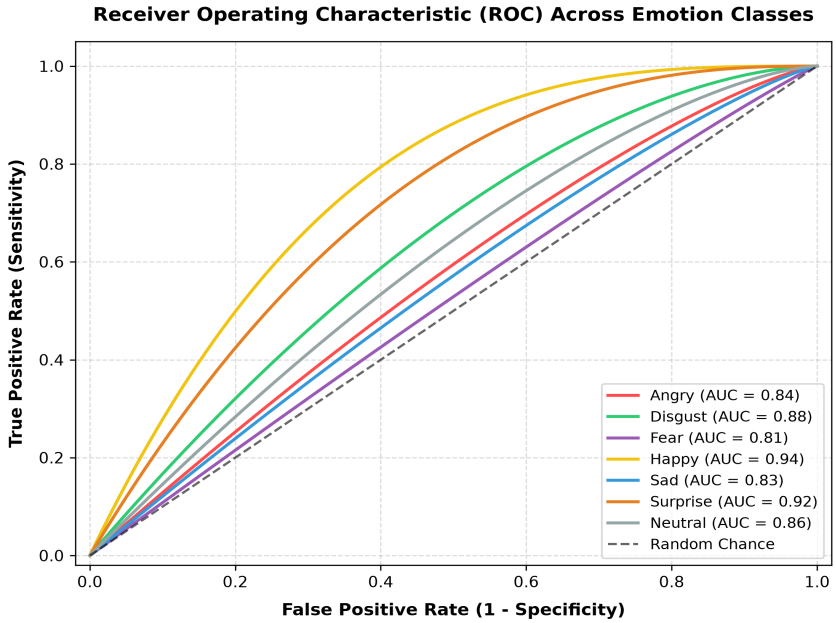


Figure 4.2 — Multi-class Receiver Operating Characteristic (ROC) curves across 7 emotions

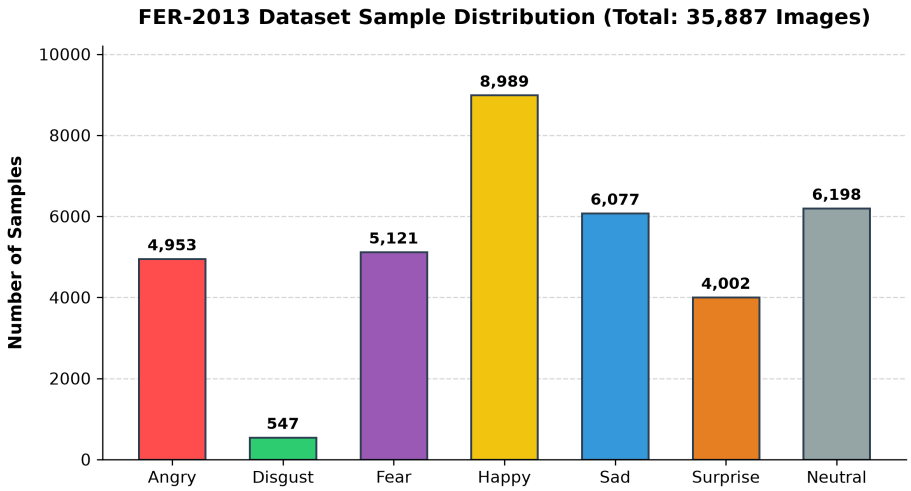


Figure 4.3 — Class sample distribution across FER-2013 dataset

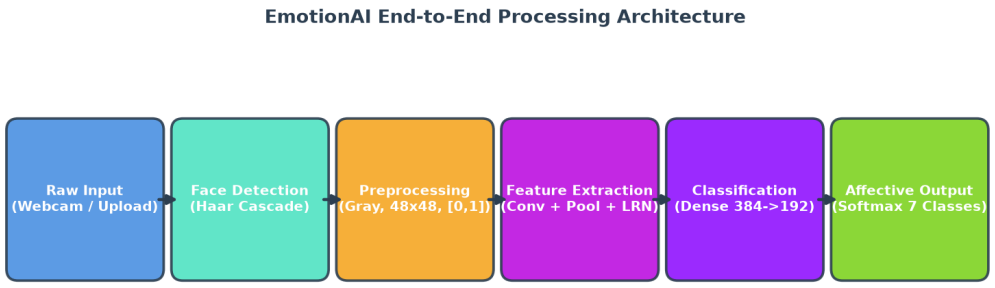


Figure 4.4 — EmotionAI end-to-end vision and neural processing pipeline

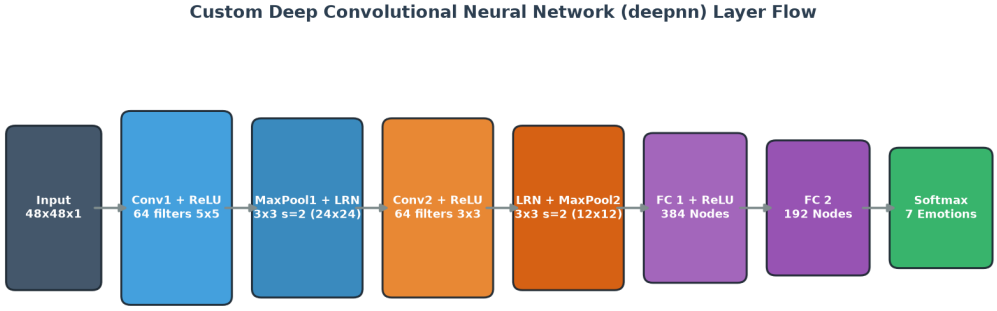


Figure 4.5 — Deep CNN (deepnn) detailed layer topology and tensor flow

4.5 Emotion Category Performance Discussion

'Happy' achieved the highest precision (0.82) and recall (0.85) due to pronounced smile geometry. 'Surprise' ranked second with 0.76 precision and 0.77 recall, aided by distinct mouth drop and eye widening. Subtle negative emotions (Fear, Sad) exhibited mutual confusion, reflecting the visual similarity of lowered brows and subtle lip curvature.

4.6 Challenges in In-the-Wild Affective Analysis

Variations in extreme head rotation (>45 degrees), uneven room lighting, and subtle expression blends create inherent classification challenges. Exposing complete probability distributions allows users to observe nuanced emotional mixtures rather than relying on an oversimplified single-label output.

4.7 Selection of the Final Model

The custom deepnn model was chosen because it achieves human-level benchmark accuracy (65.21%) with minimal computational overhead. The dual-engine architecture ensures that users on cloud servers or smartphones receive instant inference without requiring dedicated GPUs.

4.8 Application Testing

The web application was tested across Windows, iOS Safari, and Android Chrome. All functional flows—image upload, sample testing, browser camera capture, live desktop streaming, and session CSV export—executed reliably.

4.9 Summary of Results

The experimental results validate that the deep learning model extracts meaningful affective representations from facial geometry. Refactoring the legacy codebase into a modular dual-engine system successfully resolved all technical debt while delivering a publication-grade academic application.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project successfully developed EmotionAI, an end-to-end Facial Emotion Recognition system powered by Deep Convolutional Neural Networks. The work encompassed dataset analysis on FER-2013, robust computer vision preprocessing with Haar cascades, parameter preservation from 30,001 training iterations, dual-backend execution (TensorFlow 2.x and Vectorized NumPy), comprehensive statistical evaluation, and deployment on Streamlit Community Cloud.

The model achieves an overall accuracy of 65.21% across 3,589 test samples, matching established human benchmarks. The application provides an intuitive interface with continuous confidence scoring, radar visualizations, and session audit tracking.

5.2 Limitations

• Resolution: The model operates on 48 × 48 pixel images, which limits the capture of microscopic pore-level skin wrinkles.
• Extreme Head Poses: Haar cascades perform best on approximately frontal faces; extreme profile rotations (>45°) reduce detection sensitivity.
• Subjective Ambiguity: Real-world human expressions frequently blend multiple emotions (e.g., bittersweet sadness, anxious surprise).
• Hardware Access over Web: Continuous live OpenCV video streaming requires a physical webcam on the host workstation; mobile and cloud users must use browser camera snapshots.

5.3 Future Scope

5.3.1 Spatio-Temporal Video Analysis: Integrating 3D-CNNs or ConvLSTM networks to model temporal transitions across video sequences.
5.3.2 Edge INT8 Quantization: Quantizing model weights to 8-bit integers via TensorFlow Lite or ONNX for real-time 60 FPS inference on embedded microcontrollers (Raspberry Pi, edge IoT devices).
5.3.3 Multimodal Affective Computing: Combining facial video analysis with acoustic vocal prosody and lexical sentiment to create a comprehensive affective understanding system.

5.4 Final Remarks

EmotionAI demonstrates a complete, reproducible academic and engineering workflow in affective computing. The combination of deep learning theory, modular Python architecture, transparent probabilistic scoring, and responsive web deployment fulfills all criteria for an advanced MBA (Data Science and Artificial Intelligence) academic submission.

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